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The Palaeomagnetism of the Gulf of Guinea Volcanic Province, West Africa

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Summary

Palaeomagnetic results are presented for 205 sites and over 600 cores from Upper Tertiary to Recent lavas and intrusions in the Gulf of Guinea volcanic area of equatorial West Africa.

The island of Annobon is built up of a suite of reversely magnetized lavas about 18·4 My in age which give a mean direction of $D = 189{\cdot}2°$, $I = 11{\cdot}2°$ ($N = 7$, $\alpha_{95} = 15{\cdot}3°$). A younger phase of volcanism dated as 2·6 My has produced lavas with a mean direction of magnetization of $D = 337{\cdot}1°$, $I = -44{\cdot}3°$ ($N = 7$, $\alpha_{95} = 9{\cdot}3°$). The accessible parts of the island of São Tomé are covered by young lavas of Pliocene and younger age with both polarities. Forty nine significant sites from this island combine to give a mean direction of $D = 359{\cdot}2°$, $I = -7{\cdot}0°$ ($\alpha_{95} = 4{\cdot}7°$). The Island of Principe is an extensively eroded complex of lavas intruded by phonolites and tephrites believed to be Lower Miocene in age. Twenty five acceptable sites combine to give a mean direction of $D = 187{\cdot}2, I = -3{\cdot}2°$ ($\alpha_{95} = 8{\cdot}9°$). Both polarities are present in the lavas and intrusions on this island. All but two of 52 sites on Fernando Poo are normally magnetized and the bulk of this island is believed to have been built up in the Brunhes epoch. The sites combine to give a mean direction of $D = 359{\cdot}9°$, $I = -3{\cdot}0°$ ($\alpha_{95} = 4{\cdot}4°$) and the derived palaeomagnetic pole is significantly different from the present pole implying that the field here has not approximated to an axial geocentric dipole in Brunhes epoch times. Thirty-two sites distributed through the volcanic area of Cameroun combine to give a mean direction of $D = 4{\cdot}6°$, $I = 4{\cdot}3°$ ($\alpha_{95} = 6{\cdot}1°$). The sites around Mt Cameroun are all normally magnetized and this volcano is believed to have grown in Brunhes epoch times.

Analysis of the Upper Tertiary palaeomagnetic results from the African plate demonstrates no detectable movement of the plate relative to the geographic pole for about the past 25 My.

1. Introduction

The Gulf of Guinea volcanic province is a linear belt of volcanism that extends more than 300 km across central West Africa from the Bambouto and Manengouba mountains in Cameroun to Mt Cameroun and the island of Fernando Poo in the Gulf of Guinea (Fig. 1). It continues seawards for a further 700 km through the islands of Principe, São Tomé and Annobon, and may extend further south-west of

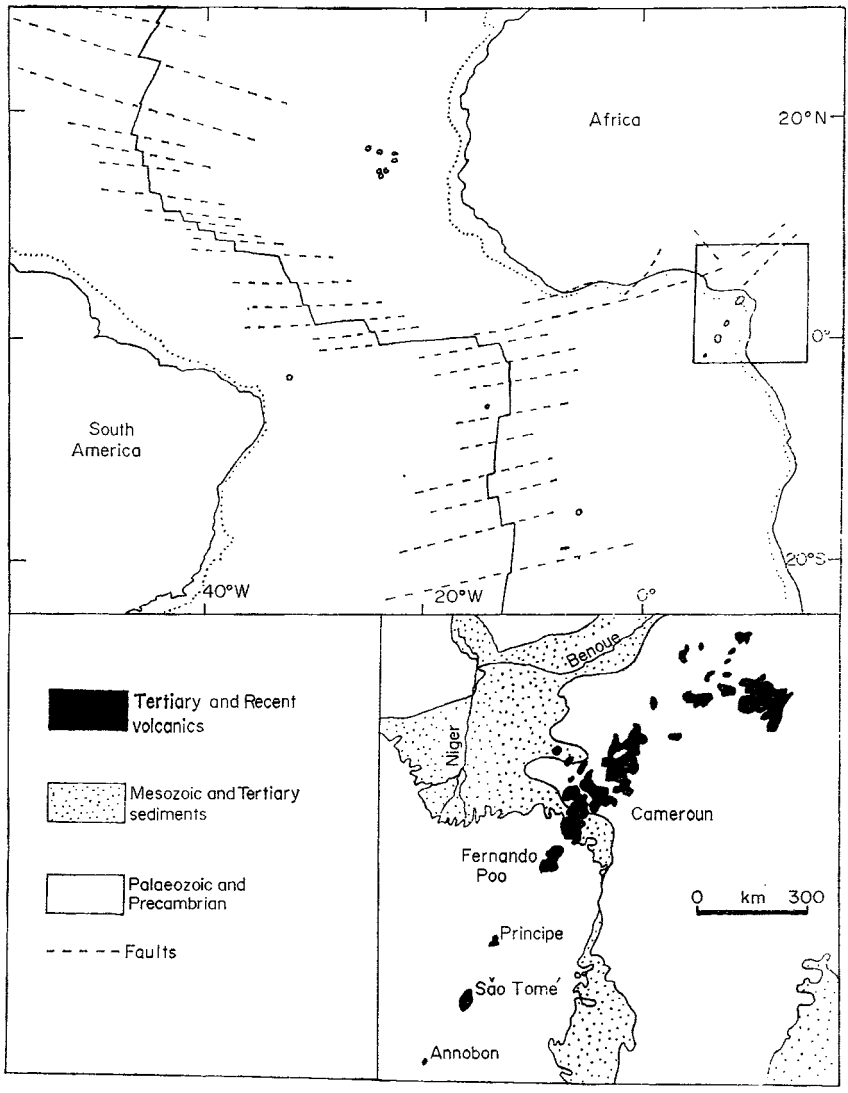


FIG. 1. Location of the Gulf of Guinea area in relation to the mid-Atlantic ridge and the faults mapped along the West African coast by Fail *et al.* (1970). The lower diagram shows the distribution of volcanic rocks in the Gulf of Guinea area.

Annobon as the Guinea submarine rise on which a number of large seamounts have been reported (Hall 1966). The north-eastward continuation of the lineament may include the volcanic fields which occur sporadically along the Massif of Adamaoua to Chad and Tibesti. The lineament thus cuts across both oceanic and continental crust and is one of major post-Palaeozoic structural elements of the African continent. This paper presents the results of a palaeomagnetic study of the volcanic rocks of the islands in the Gulf of Guinea and the south-western part of the volcanic line in Cameroun (Piper 1970).

2. Geological outline

2.1 *Regional setting*

According to Wright (1968) the opening of the South Atlantic progressed from the south in Jurassic times and reached the Gulf of Guinea area by the Mid-Cretaceous. Grant (1971) favours a more or less synchronous opening along the entire common margin in Aptian times (100–106 My). Sedimentation in the Gulf of Guinea area began with the deposition of coarse sandstones in Aptian times in south-east Nigeria. The Gulf area developed as an unstable Rift-Rift-Fracture junction (McKenzie & Morgan; Grant 1971) for approximately 30 My until Santonian times, during which the Benue trough of Nigeria opened. Strains accumulated within the African plate were released after the cessation of this opening, and partial recovery led to the folding of sediments in the Benue trough in Santonian times (76–82 My). Sedimentation in the Cameroun area continued uninterrupted to lower Eocene times, after which there is a gap in the stratigraphic record which appears to coincide with the development of the Massif of Adamaoua as a faulted monoclinal margin to the Benue rift. The faulting and uplift of this massif coincides with the first phase of volcanism in the Bamenda Highlands of Cameroun (Géze 1953) but volcanism did not commence in the Mt Cameroun area until middle Miocene to early Pliocene times (Hedburg 1968).

The volcanic islands are distinctly oceanic in chemical character and are formed principally of alkali olivine basalts and basic accumulates; the lavas are commonly intruded on the islands of Principe, São Tomé and Annobon by domes of trachyte, phonolite and tephrite. Mt Cameroun is a basaltic shield volcano which is morphologically and chemically similar to the volcanoes of Fernando Poo, but north-eastwards along the lineament rocks of calc-alkaline affinities become more important. Andesites, peralkaline trachytes and rhyolites are found in the Bambouto, Roumpi and Maenengouba mountains (Géze 1943, 1953; Jeremine 1941, 1943). The Guinea volcanic province has had a complex history and several magmatic episodes are represented. Alkali igneous intrusions in Cameroun give Lower Tertiary ages (Cahen & Snelling 1966; Hedburg 1968) and more than one volcanic phase is present on all islands except Fernando Poo. The volcanoes of Mt Cameroun and Santa Isabel on Fernando Poo are still active. The volcanic area in Cameroun is paralleled by a similar but more restricted sequence on the Jos Plateau on the north-western flank of the Benue trough.

2.2 *Annobon*

This is a strongly-cliffed volcano built up of alkali olivine basalts rich in ferromagnesian minerals and often containing olivine nodules. Three phases of volcanism are represented on the island; the first involved the construction of the main volcanic edifice which was subsequently intruded to high levels by domes of trachyte (Fig. 2). After partial erosion a small amount of renewed volcanism took place from the north-western rim of the crater and lavas flowed down to the north coast. Schultze (1913) has presented a topographic map of the island and includes some geological notes. Tyrell (1934) has described the petrology of some rocks from Annobon, and several chemical analyses are presented and discussed by Fuster Casas (1950).

2.3 *São Tomé*

São Tomé is built up largely of alkali olivine basalts but phonolites, andesites and leucite-basanites also occur. The centre of the island is a rugged and heavily eroded volcanic complex intruded by phonolite and tephrite domes. It is largely inaccessible and has not so far been mapped (Fig. 3). The bulk of the north-east and southern

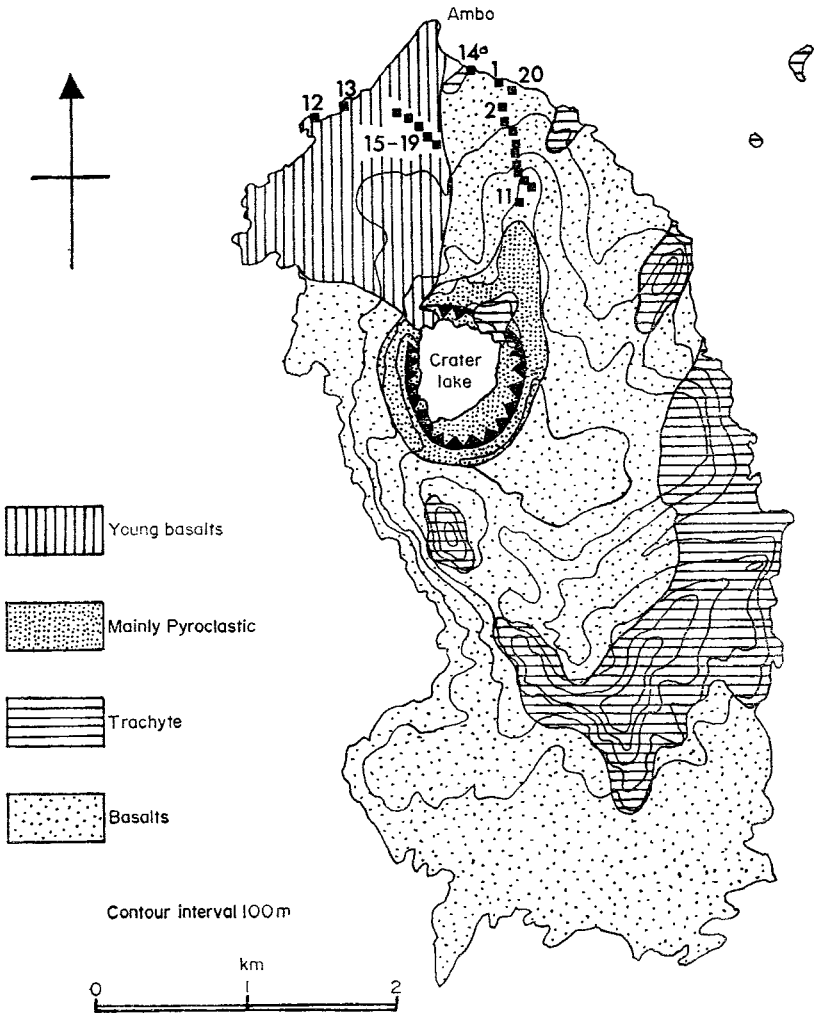


FIG. 2. Geology and sampling sites on the island of Annobon in the Gulf of Guinea (5·5° E, 1° S.).

parts of the island are covered by young basalt lavas, and a number of recent basalt scoria cones are found at the north-east and southern ends of the island and on the Rolas islet. Below these young basalts in the centre of the island a 'window' of sandstone (the Ubabudo Sandstone) is exposed, and although this contains no completely diagnostic fossils it is believed to have been deposited in Lower to Upper Cretaceone times (Hedburg 1968). This sandstone, which has moderate to steep dips to the north, contains metamorphic minerals, and suggests that the island may incorporate a continental fragment from the initial separation of Africa and South America. The sedimentary rocks have been described by Teixiera (1949, 1955) and the petrology of the volcanic rocks has been discussed by Neiva (1954a, b, 1956a), Neiva & Ribiero de Albuquerque (1962) and Assunçao (1956, 1957). Some aspects of the geology are discussed by Neiva (1956b), Pereira (1944), and Neiva & Ribeiro de Albuqueque (1962).

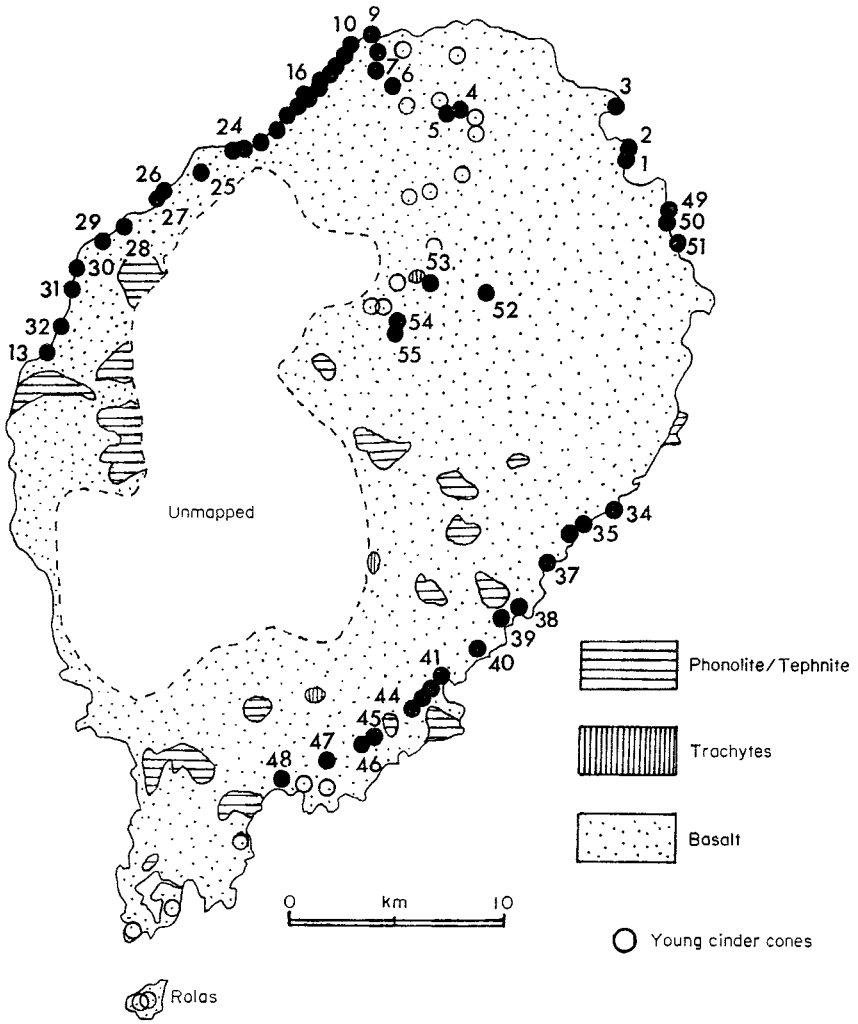


FIG. 3. Geology and palaeomagnetic sampling sites on the island of São Tomé (6·5° E, 0° S).

2.4 *Principe*

The island of Principe is the remnant of a much larger volcano which is defined by the extensive surrounding submarine shelf and a number of islets which occur in the vicinity. At the present level of erosion large intrusive bodies of tephrite and phonolite form a continuous outcrop across most of the southern half of the island. In the north of the island they intrude the remaining lava pile as domes (Fig. 4). The lavas are alkali olivine basalts sometimes tending towards mugearites. They are frequently heavily lateritized and much of the plateau in the north of the island above 100 m is composed of a thick laterite deposit. Miocene limestones from the island have been described by Da Silva (1958) but doubt has been cast on these by Hedburg (1968), and the igneous petrology has been studied by Barros (1960) and Neiva (1954b, 1955a, b, 1956b).

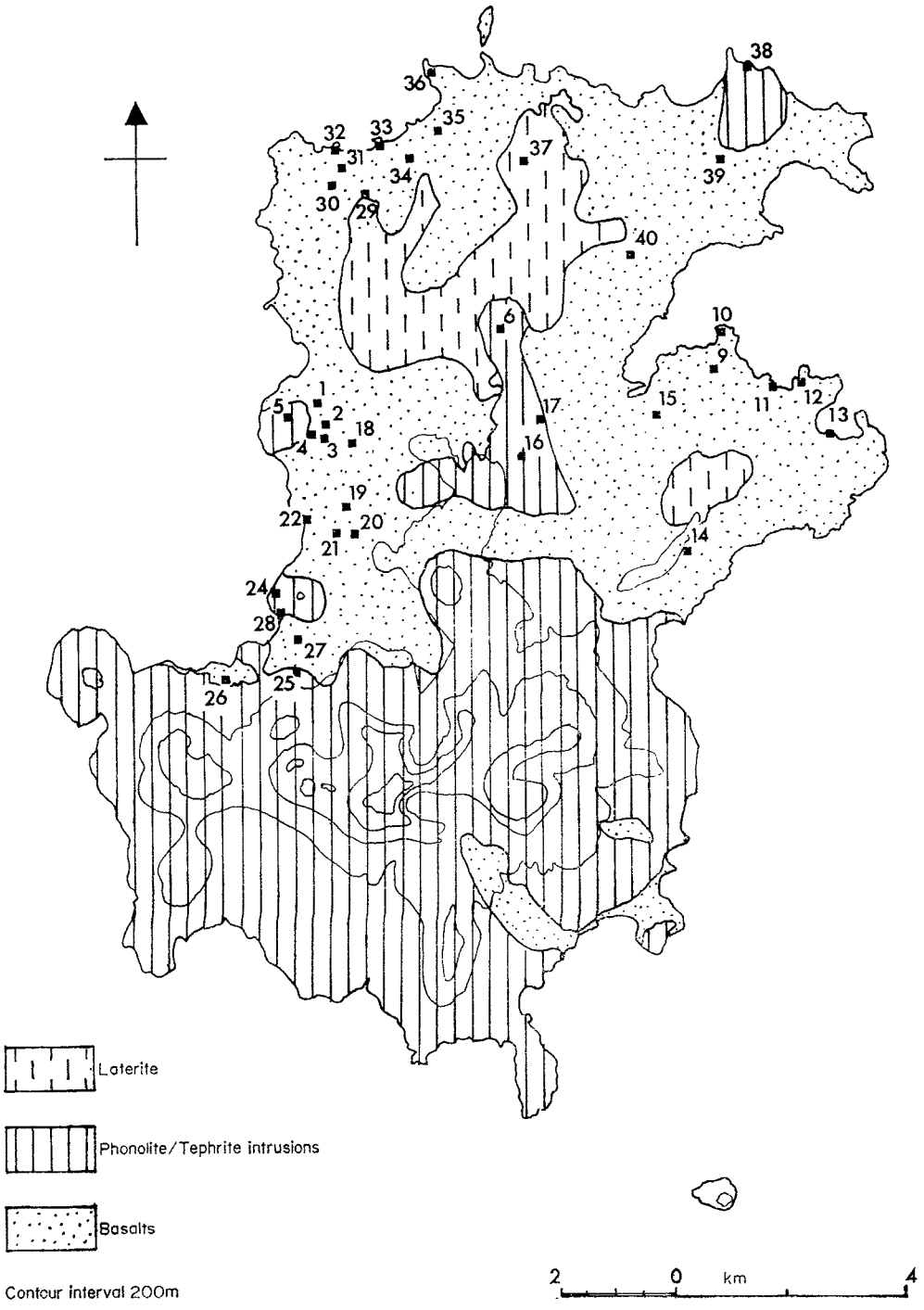


FIG. 4. Geology and palaeomagnetic sampling sites on the island of Principe (7·5° E, 1·5° N).

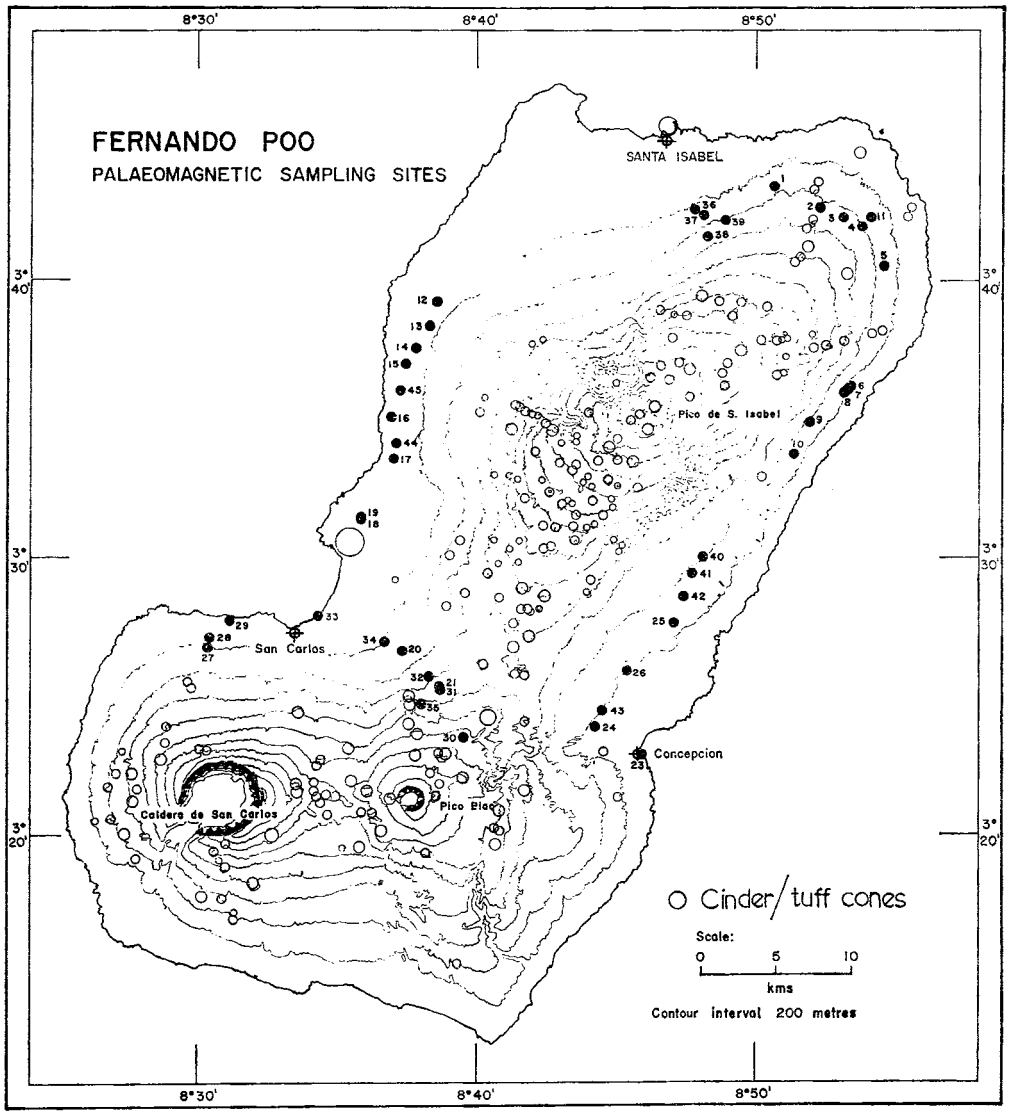


FIG. 5. Geology and palaeomagnetic sampling sites on the island of Fernando Poo.

2.5 *Fernando Poo*

Fernando Poo is a youthful island with a topography that is entirely constructional. The northern most volcano (Santa Isabel) lies on the continental shelf and young flows have been emptied from vents on the main fissure zone which is marked by numerous small cinder running along the crest of the volcano (Fig. 5). An eruption was recorded on the south-east slope of this volcano in 1898 and many of the lava flows are evidently very young. The southern two volcanoes (San Carlos, Pico Baio) straddle the continental slope and they are covered by many small vents but show no marked linearity. All the rocks on this island are alkali olivine basalt flows often approaching picrites, oceanites, or ankaramites in composition (Fuster Casas 1956).

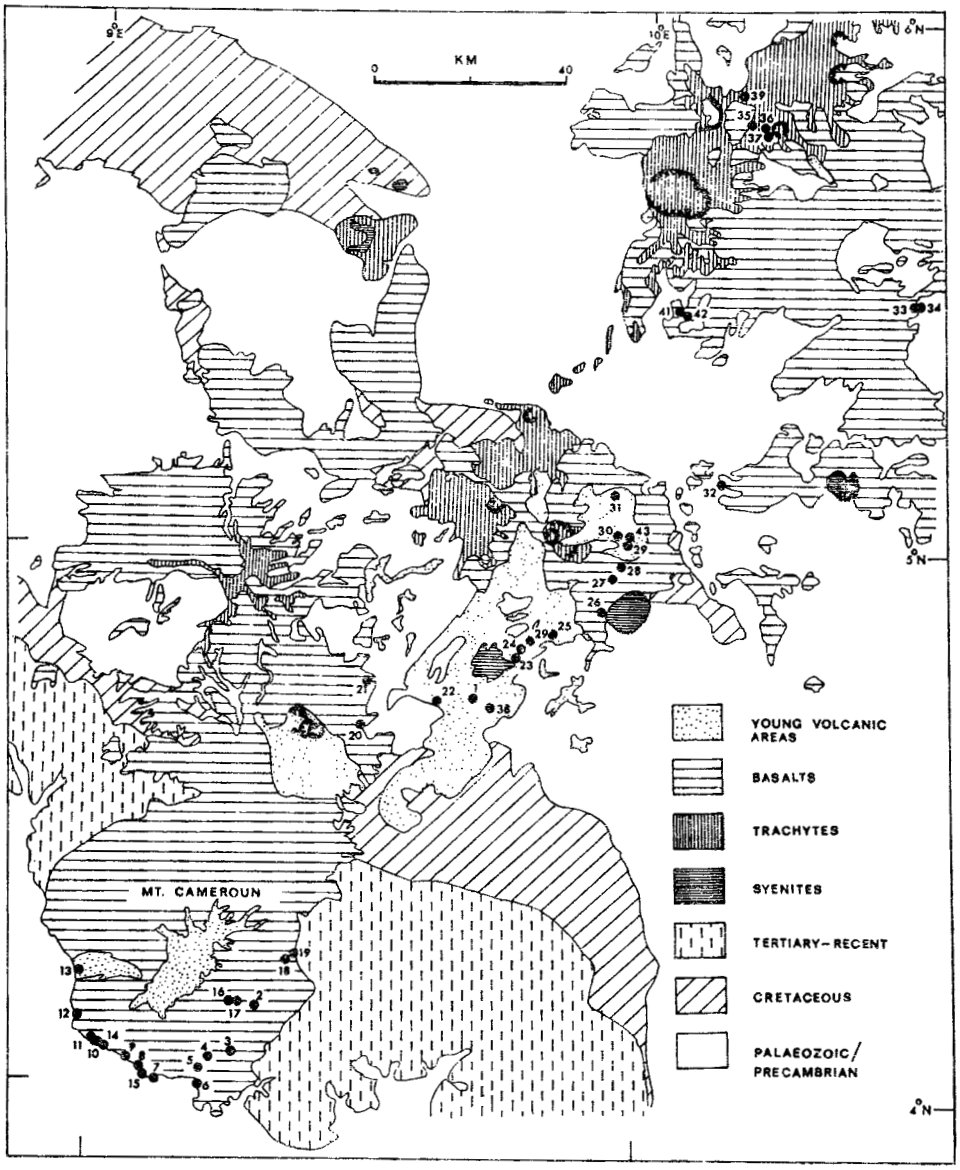


FIG. 6. Geology and palaeomagnetic sampling sites in West Cameroun.

2.6 *Cameroun*

The volcanics of Cameroun belong to three groups (Géze 1953). The oldest is a series of basalt and andesite flows which are now heavily laterized but appear to have come from fissure eruptions following the regional SSW–NNE trend and may be Lower Tertiary in age. These rocks cover extensive areas between Mt Cameroun, the Roumpi Mountains and the Manengouba Massif, and occur again in the Bamenda Highlands (Géze 1943). The middle sequence is represented by a series of trachyte and phonolite complex composed of lavas intruded by domes and plugs, in which calderas are frequently developed (Fig. 6). This sequence reaches a considerable thickness in parts of the Bamenda Highlands but dies out south-westwards.

The upper group is represented by Mt Cameroun and a field of numerous small cinder cores and young lava flows which runs for 100 km to the north-east. Mt Cameroun is one of the most active volcanoes in the world, 13 eruptions having been reported since 1800 (Guillaume 1966); most of the recent activity has occured along the crest of the volcano which is marked by numerous small cinder cones and small dilation cracks (Fig. 6).

3. Results

3.1 *Field and laboratory methods*

Up to five samples were collected using a portable drill of the type described by Doell & Cox (1967) from each of 158 sites on São Tomé, Principe, Fernando Poo and Cameroun. On Annobon lump samples were collected, and since time was only available for one or two samples to be collected from each site the results cannot be expected to be of the same quality as for the other areas. Wherever possible the samples were orientated with respect to geographic north by sights on the Sun or features of known position. When this was not possible orientations were made by compass and corrected for the mean declination at the site. Sampling was controlled by the limited exposure and inaccessibility of much of the area, and was generally confined to roadcuts and coastal exposures. Owing to the vegetation cover it was generally not possible to establish whether each site was on a different rock unit but it is considered that the areas sampled are sufficiently large to keep duplication down to an acceptable minimum.

The magnetism of the samples from Cameroun and Fernando Poo was measured with an astatic magnetometer, and for the samples from the other localities with a spinner magnetometer (Doell & Cox 1965); the susceptibilities were measured with a bridge system of the kind described by Collinson & Molyneux (1967). The magnetic properties of the collections are summarized in Table 1 and normal probability curves fitted to the distributions of magnetic properties are shown in Fig. 7. The highest intensities are associated with the collection from Fernando Poo which consists of basalt lavas belonging largely to the Bruhnes epoch. The collections from São Tomé and Cameroun which include older lavas are somewhat weaker. The collection from Principe contains a large percentage of phonolites and with weaker intensities than typical basalts.

The samples were subjected to progressive demagnetization in alternating fields up to 300 or 400 Oe in steps of 100 Oe. The mean cleaned direction for each site was computed using a least scatter criterion in which the cleaned directions are combined

Table 1

Summary of mean magnetic properties of the rock collections from the Gulf of Guinea volcanic province. Intensities and susceptibilities and Q ratios are log. normal means. The Stability index S200 is defined by Wilson et al. (1968). *Q200 is the Q ratio after demagnetizing in fields of* 200 Oe.

	Magnetic susceptibility (Gauss/Oe)	Intensity of magnetization (Gauss)	Stability index S200	*Qo*	*Q200*
Annobon	0·0022	0·0080	0·2639	9·954	1·302
São Tomé	0·0028	0·0053	0·2739	5·220	**1·441**
Principe	0·0028	0·0044	0·1923	11·317	**1·219**
Fernando Poo	0·0024	0·0115	0·2025	15·242	**3·174**
Cameroun	0·0023	0·0073	0·2816	**9·905**	**3·885**

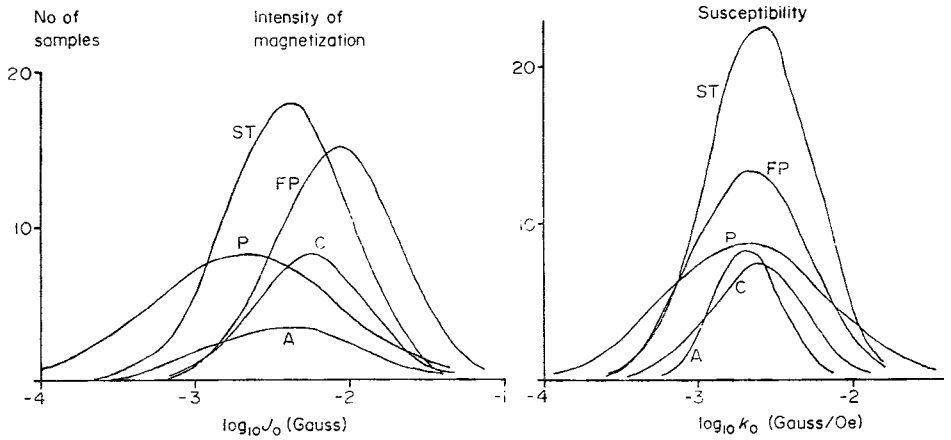


FIG. 7. Normal probability curves fitted to the distribution of magnetic susceptibilities and intensities for the rock collections from the Gulf of Guinea area. The collections are: Annobon (A), São Tomé (ST), Principe (P), Fernando Poo (FP) and Cameroun (C)

in different combinations until a maximum precision (Fisher 1953) is achieved (Watkins & Richardson 1968). A significance test (Irving 1964, page 64) has been applied and sites whose directions are not significantly different from random at the 95 per cent confidence level have been omitted from the subsequent analysis. The most serious problem encountered during the magnetic cleaning of these collections was that of large IRMs which have characteristics suggesting that they were acquired by lightning strikes (Graham 1961). Without exception the sites rejected have large secondary components attributable to this cause.

The mean directions of magnetization for sites from each sampling area are plotted in Fig. 8 and the statistics are given in Table 2. The mean palaeomagnetic pole positions and 95 per cent confidence circles have been computed for the four islands and Cameroun by assigning unit weight to each significant site mean direction using Fisher's (1953) statistics.

3.2 *Polarity variations*

In this sub-section the geomagnetic polarity changes observed in each locality will be considered in relation to present knowledge of the age and volcanic history of the islands.

3.2.1 *Annobon.* The sites 1, 2–11 and 20 are distributed through the volcanic pile in the northern part of this island to the base of the predominantly pyroclastic succession observed around the crater (Fig. 2). All these sites except for sites 1 and 9 have a reverse magnetization. The basal lava of this succession has given K–Ar ages of 19·2 and 17·5 My with a preferred average of 18·4 My. Site 9 has a near-normal declination but an anomalous inclination and may record an incomplete reversal of the magnetic field. If the anomalous sites 1 and 9 are rejected the remaining seven sites combine to give a mean reversed direction of $D = 189{\cdot}2°$, $I = 11{\cdot}2°$. Site 14 was taken from an intrusive trachyte plug and like site 9 has a normal declination but a steep inclination. Sites 12, 13, and 15–19, were taken from the group of young lavas which have flowed down from the crater rim to the north coast. It is not known how many units are present here but they may represent one eruptive episode only. All sites are normal with a steep inclination and combine to give a mean direction of $D = 337{\cdot}1°$, $I = -44{\cdot}3°$ (Table 3) and the derived palaeomagnetic pole is significantly different from the present pole (Fig. 8). A sample of this lava has

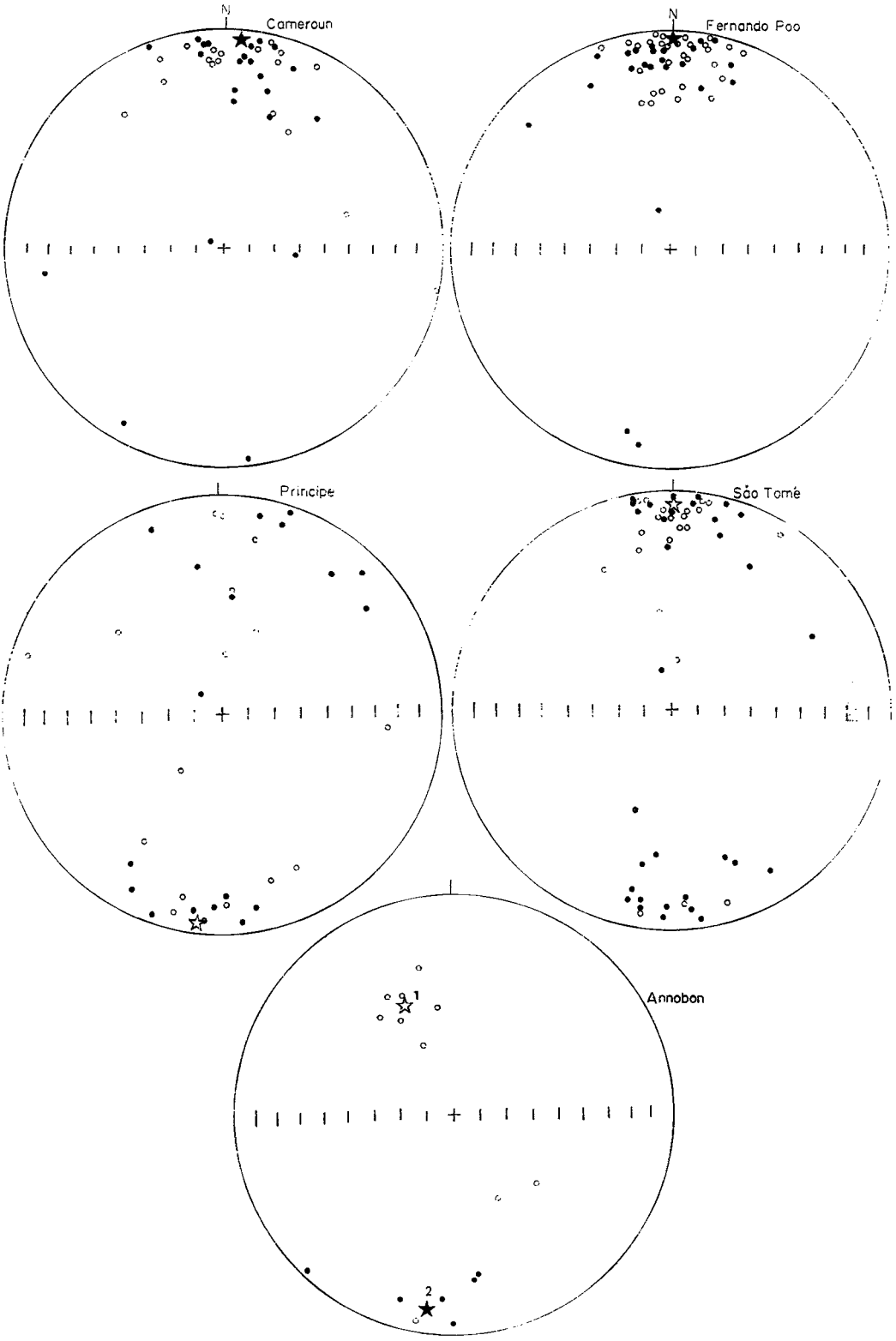


FIG. 8. Site mean directions of magnetization for the collections from the Gulf of Guinea volcanic province. The stars are mean directions of magnetization for the acceptable sites (see Tables II and III).

Table 2

Gulf of Guinea volcanic province, palaeomagnetic data

(a) *Cameroun*

Site number	Type	Number of samples	D	I	k_{100}	k_{200}	k_{max}	Palaeomagnetic pole °N ϕ'	°E θ	α_{95}	Polarity
1	L (B)	5	350	−4	117·4	5·0	124·4	77·6	235·8	6	N
2	L (B)	3	340	19	157·1	126·3	540·5	70·0	289·8	5	N
3	L (B)	3	12	−3	723·7	4416·8	5031·8	75·8	125·7	1	N
4	L (B)	3	2	26	15·5	20·0	29·3	80·9	22·6	23	N
5	L (B)	2	13	3	27·4	2·6	27·4	69·5	109·5	24	N
6	L (B)	2	11	19	92·5	14563·6	14563·6	66·2	287·6	44	N
7	L (B)	3	262	17	0·9	1·0	1·0	7·9	287·9	44	N
8	L (B)	2	5	11	401·9	520·9	1568·0	56·9	78·3	32	N
9	L (B)	2	20	10	12·6	7·4	13·0	69·4	95·3	76	N
10	L (B)	1	173	3	—	—	—	62·1	71·8	44	N
11	L (B)	3	303	83	1·5	1·4	1·5	67·4	331·1	25	N
12	L (B)	3	7	6	442·0	2967·2	4730·6	81·8	112·6	1	N
13	L (B)	3	341	−8	12·4	10·8	13·6	69·0	248·9	34	N
14	L (B)	2	27	−6	60·7	96·4	97·7	70·0	124·6	29	N
15	L (B)	3	355	6	13·7	12·9	16·1	84·9	246·7	31	N
16	L (B)	2	35	−26	2·1	9·7	14·6	69·6	119·1	79	N
17	L (B)	3	7	−7	415·3	9·1	470·0	78·2	147·8	5	N
18	L (B)	3	8	22	115·2	417·5	794·4	79·9	58·2	4	N
19	L (B)	4	356	26	255·0	157·4	327·1	80·9	145·5	5	N
20	L (B)	3	8	13	79·6	149·9	310·9	82·0	86·4	7	N
21	L (B)	5	183	−3	3·4	4·0	4·7	84·9	326·1	39	R
22	L (B)	3	9	−7	132·9	157·3	157·3	76·6	138·8	9	N
23	L (B)	2	15	25	1·9	1·9	2·0	67·0	2·7	87	N
24	L (B)	3	3	31	78·9	5·7	109·2	77·8	21·6	11	N
25	L (B)	2	9	3	1·4	1·3	1·5	80·0	117·4	43	N
26	L (B)	3	354	5	84·9	22·6	84·9	83·5	248·5	13	N
27	L (B)	3	355	−11	75·5	31·9	97·4	77·7	208·6	12	N
28	L (B)	2	355	−11	4·2	3·0	4·5	59·5	287·3	66	N
29	L (B)	3	358	−8	25·6	73·2	74·5	79·9	195·0	14	N
30	L (B)	3	18	34	2·8	4·5	11·2	87·0	44·0	38	N
31	L (B)	3	356	−6	4·9	207·8	259·7	80·7	208·8	7	N

32	L (B)	5	210	9	22·4	14·2	35·3	57·9	293·4	13	R
33	L (B)	5	352	2	63·7	145·7	149·2	81·7	244·4	6	N
34	L (B)	4	323	−23	6·0	54·8	69·3	49·4	249·2	11	N
35	L (T)	5	19	−34	1·1	1·0	1·2	59·3	148·2	42	N
36	L (T)	3	103	−62	9·6	8·4	9·6	13·5	139·6	42	—
37	L (T)	3	74	−41	19·6	22·6	22·6	11·6	121·4	55	—
38	L (B)	3	4	13	131·3	112·7	112·7	85·7	74·6	9	N
39	L (B)	3	29	−39	122·6	17·2	17·2	49·6	140·7	2	N
42	L (B)	3	339	—	14·5	20·3	20·3	69·2	261·1	21	N
43	L (B)	1	15	−5	—	—	—	72·6	122·6	—	N
					(b) *Fernando Poo*						
1	L (B)	4	359	−6	97·3	318·5	348·9	83·4	193·3	4	N
2	L (B)	4	6	7	26·3	34·9	41·8	83·2	97·9	14	N
3	L (B)	3	10	25	8510·9	77·3	8510·9	75·2	55·0	2	N
4	L (B)	3	347	16	224·2	321·8	926·9	76·9	360·9	4	N
5	L (B)	4	325	15	21·9	277·1	316·0	81·4	308·6	5	N
6	L (B)	4	4	−2	164·9	210·9	218·2	83·4	148·3	6	N
7	L (B)	4	355	9	8·2	13·8	23·1	84·8	293·1	19	N
8	L (B)	3	333	17	4·7	2·4	7·8	63·4	292·1	49	N
9	L (B)	2	354	16	153·9	230·8	230·8	82·3	318·7	17	N
10	L (B)	3	354	−28	122·7	82·3	122·7	70·6	205·8	12	N
11	L (B)	1	21	19	—	—	—	67·9	80·4	—	N
12	L (B)	3	5	−3	14·0	79·6	137·7	82·1	141·6	10	N
13	L (B)	4	20	−3	18·5	36·2	44·3	68·7	113·0	14	N
14	L (B)	5	356	−0	33·8	29·4	35·3	84·8	233·5	13	N
15	L (B)	5	11	−0	132·5	124·4	273·8	78·4	117·9	4	N
16	L (B)	3	348	8	157·9	153·4	195·0	77·9	281·8	8	N
17	L (B)	4	4	−25	4·1	3·1	5·0	72·4	175·5	49	N
18	L (B)	4	3	−31	116·5	347·6	1221·8	69·3	180·7	2	N
19	L (B)	3	14	−13	43·0	75·2	120·1	72·8	135·0	11	N
20	L (B)	3	354	6	5·0	23·0	28·7	84·3	276·2	23	N
21	L (B)	3	340	−2	46·8	16·1	68·9	69·9	265·0	14	N
22	L (B)	3	359	5	145·3	173·5	686·2	89·2	213·8	4	N
23	L (B)	3	7	−3	1·6	1·9	255·0	81·3	135·6	8	N
24	L (B)	3	340	−1	1·9	1·9	2·0	70·2	267·4	41	N
25	L (B)	4	325	−33	31·9	69·2	11·8	67·2	208·0	9	N
26	L (B)	3	10	−4	110·6	91·8	169·3	78·2	126·7	9	N

Site number	Type	Number of samples	D	I	k_{100}	k_{200}	k_{max}	Palaeomagnetic pole °N ϕ'	°E θ'	α_{95}'	Polarity
27	L (B)	3	359	−15	374·9	40·0	625·5	78·8	193·4	5	N
28	L (B)	3	358	−5	338·4	17·4	338·4	83·3	205·8	6	N
29	L (B)	4	358	9	175·3	290·8	290·8	87·8	320·5	5	N
30	L (B)	2	358	−14	6033·3	87·4	6033·3	79·2	195·3	3	N
31	L (B)	4	4	−10	168·5	154·3	217·2	80·4	164·1	6	N
32	L (B)	3	2	−5	897·6	118·7	3964·1	83·3	171·4	1	N
33	L (B)	3	339	5	1·9	1·9	2·2	69·9	277·2	58	N
34	L (B)	2	311	14	2·5	3·9	3·9	41·9	285·6	39	N
35	L (B)	3	18	11	20·5	16·4	59·1	71·0	91·4	16	N
36	L (B)	6	15	−28	105·4	121·1	169·9	66·1	150·7	33	N
37	L (B)	3	16	−1	3·8	4·8	4·9	73·2	112·4	8	N
38	L (B)	3	4	15	126·9	274·1	372·1	84·0	55·5	6	N
39	L (B)	4	358	−1	642·8	327·3	1702·0	85·5	216·4	2	N
40	L (B)	3	351	−5	47·7	4·6	49·1	79·1	244·1	17	N
41	L (B)	4	350	−16	18·5	10·1	21·4	74·3	228·4	20	N
42	L (B)	4	348	−32	849·7	166·4	2260·7	66·4	216·7	1	N
43	L (B)	3	11	−6	31·2	17·8	65·4	77·1	130·3	15	N
44	L (B)	3	357	−27	325·2	267·5	377·8	71·6	198·5	6	N
45	L (B)	3	355	−2	528·2	186·6	528·2	83·0	234·3	5	N
47	L (B)	3	358	15	136·9	1·9	182·9	81·8	350·0	9	N
49	L (B)	4	349	0	486·3	168·2	865·1	79·0	278·1	3	N
50	L (B)	3	192	15	0·8	1·5	1·9	75·6	310·0	41	R
51	L (B)	4	188	9	2·9	97·5	97·5	80·3	306·5	9	R
52	L (B)	3	8	3	10·9	12·1	58·9	81·7	85·0	16	N
54	L (B)	3	349	−3	61·4	180·7	180·7	79·5	267·0	9	N
55	L (B)	3	350	7	67·4	54·6	69·7	79·6	299·2	14	N
					(c) *Principe*						
1	L (B)	3	179	18	68·3	3·6	68·3	79·1	12·3	15	R
2	L (B)	4	45	9	4·2	1·7	7·7	45·5	92·5	34	—
3	L (B)	2	5	44	5·6	5·6	8·9	64·8	18·3	17	N
6	I (P)	3	314	78	99·1	15·6	172·5	16·8	351·3	1	—
8	I (P)	3	339	9	2·9	1·4	3·8	69·3	286·9	34	N
9	L (B)	3	351	31	4·3	4·3	4·4	72·4	339·8	9	N
10	L (B)	3	212	21	36·3	9·1	74·3	55·0	299·9	12	R

11	L (B)	4	185	−6	42·3	79·2	92·4	84·0	262·5	5	R
12	L (B)	4	164	−23	186·1	84·7	300·5	71·1	133·1	8	R
13	L (B)	4	208	11	87·7	24·5	111·4	60·5	292·2	6	R
14	L (B)	3	54	19	89·9	79·7	125·8	35·5	86·4	11	—
15	I (P)	3	213	−32	248·1	141·1	401·1	53·7	249·2	6	R
16	I (P)	2	200	4	3·4	3·3	5·1	69·5	288·4	21	R
17	I (P)	3	3	−66	2·5	3·4	3·4	39·4	184·7	14	—
18	L (B)	3	39	19	5·3	11·9	37·5	49·9	83·7	14	N
19	L (B)	2	195	−8	1·7	2·0	2·4	74·3	267·8	54	R
20	L (B)	2	189	11	1·0	1·1	8·2	78·1	315·7	25	R
21	L (B)	2	179	14	25·2	62·1	57·9	81·0	12·1	9	R
23	I (T)	5	19	1	57·6	81·7	116·6	70·0	99·8	7	N
24	I (T)	3	22	−55	27·0	10·8	28·2	47·0	160·9	23	—
25	I (P)	3	154	−23	1·2	1·0	2·0	62·2	121·7	10	R
26	I (P)	2	300	−37	292·8	462·9	462·9	27·4	252·8	11	—
27	I (P)	2	11	−19	1·2	0·9	1·7	73·6	141·8	50	N
28	I (P)	4	359	−7	175·0	31·2	225·2	84·7	195·6	6	N
29	L (B)	3	18	8	680·9	14·4	680·9	71·7	89·1	4	N
30	L (B)	3	11	7	133·6	165·8	364·8	78·2	86·2	6	N
31	L (B)	4	171	11	42·6	43·1	85·1	78·9	56·8	8	R
32	L (B)	4	182	13	5·1	13·5	61·2	81·5	349·5	18	R
33	L (B)	3	175	5	1·0	1·2	52·7	83·6	58·2	17	R
34	L (B)	2	218	−63	2·4	78·0	78·0	34·9	219·8	28	—
35	L (B)	3	5	−41	2·6	1·1	3·8	63·8	175·6	26	N
36	L (B)	3	192	−16	7·6	14·1	14·1	76·3	247·6	12	R
37	L (B)	3	0	−8	5·1	4·6	12·9	84·2	186·8	8	N
39	L (B)	3	95	−24	17·5	12·5	105·3	5·5	110·5	12	—
40	L (B)	3	288	−6	2·4	2·7	3·0	17·9	273·5	49	—

(d) *São Tomé*

1	L (B)	2	12	10	28·6	134·6	612·8	76·6	73·1	10	N
2	L (B)	4	9	−3	105·2	9·9	156·6	80·5	106·0	7	N
3	L (B)	4	2	−10	58·9	9·6	75·7	84·2	160·2	10	N
4	L (B)	5	14	18	6·7	6·5	8·8	72·8	62·7	27	N
5	L (B)	3	358	24	29·7	17·5	43·5	76·9	0·6	18	N
6	L (B)	4	349	3	2·5	14·6	27·8	79·1	286·7	17	N
7	L (B)	3	200	−49	25·4	20·3	85·1	53·6	217·6	13	12R
8	L (B)	3	173	9	319·8	218·2	630·9	82·0	58·6	4	R
9	L (B)	4	175	14	766·1	12·6	766·1	81·5	36·8	3	R

Site number	Type	Number of samples	D	I	k_{100}	k_{200}	k_{max}	Palaeomagnetic pole °N ϕ'	Palaeomagnetic pole °E θ'	α_{95}	Polarity
10	L (B)	5	188	−7	17·9	21·4	24·5	81·1	252·6	15	R
11	L (B)	3	163	−9	904·9	395·9	3686·9	72·5	113·0	2	R
12	L (B)	3	8	−1	176·3	148·2	207·4	81·9	101·1	8	N
13	L (B)	3	181	5	166·3	559·9	1722·4	86·9	335·2	2	R
14	L (B)	2	62	28	0·6	0·9	1·0	26·5	79·8	44	—
15	L (B)	1	7	−7	—	—	—	81·2	122·4	—	N
16	L (B)	5	159	29	217·9	219·9	340·5	64·7	57·7	4	R
17	L (B)	5	175	−14	8·6	7·6	11·6	81·6	155·9	23	R
18	L (B)	4	150	26	694·8	63·8	694·9	58·1	69·6	3	R
19	L (B)	4	188	13	45·3	12·0	48·8	79·2	315·5	13	R
20	L (B)	2	192	16	6051·8	5160·6	9654·0	75·3	312·7	1	R
21	L (B)	4	186	33	42·7	52·4	56·1	70·6	349·4	12	R
22	L (B)	3	171	1	146·1	242·1	935·6	81·5	91·4	4	R
23	L (B)	4	190	29	155·2	8·5	204·3	71·2	335·4	6	R
24	L (B)	3	350	−52	309·0	66·5	309·0	56·1	200·6	7	N
25	L (B)	4	31	−5	28·0	35·6	38·5	58·1	101·9	15	N
26	L (B)	4	3	3	11·4	206·1	252·5	84·0	80·1	5	N
27	L (B)	3	157	25	31·4	25·2	70·2	63·8	65·5	14	R
28	L (B)	3	359	−11	64·8	301·4	1301·4	84·0	196·2	3	N
29	L (B)	3	6	0	149·1	335·9	653·8	83·7	94·8	4	N
30	L (B)	4	356	−11	254·4	489·1	891·4	82·7	220·3	3	N
31	L (B)	3	4	−16	778·2	80·1	778·2	80·5	159·5	4	N
32	L (B)	4	0	1	159·5	2·8	168·9	89·2	6·9	7	N
34	L (B)	3	14	0	274·3	15·3	319·8	75·7	94·9	6	N
35	L (B)	3	180	12	219·8	311·6	2290·8	83·5	5·1	2	R
36	L (B)	3	352	−3	143·7	711·6	1677·9	81·8	265·8	3	N
37	L (B)	3	349	−18	136·4	69·8	296·9	75·7	233·2	7	N
38	L (B)	3	356	−8	76·9	199·2	441·6	84·5	227·7	5	N
39	L (B)	2	358	−22	69·1	71·9	95·6	78·3	193·7	25	N
40	I (P)	3	2	−16	247·7	30·5	247·7	80·9	167·9	7	N
41	L (B)	3	347	−25	144·4	21·5	1124·7	71·5	229·4	3	N
42	L (B)	3	0	9	699·3	1002·1	1362·4	85·0	6·8	2	N
43	L (B)	4	147	14	21·5	22·9	51·2	57·1	82·6	12	R
44	I (P)	3	4	−8	12·8	8·9	15·8	83·5	139·9	32	N

45	L (B)	2	356	11	393·9	1554·4	1554·4	83·2	336·8	6	N
46	L (B)	4	332	−39	68·4	36·0	76·9	55·3	234·6	10	N
47	L (B)	3	353	3	6·8	1·8	8·6	83·2	293·1	33	N
49	L (B)	4	349	—	486·2	168·2	865·1	79·0	278·1	2	N
50	L (B)	3	192	12	0·8	1·5	1·9	76·2	303·1	39	R
51	I (P)	4	188	9	2·9	97·5	97·5	80·3	366·5	9	R
52	L(B)	3	19	4	10·9	12·1	12·6	69·9	89·4	36	N
53	L (B)	3	345	74	12·8	3·4	4·9	27·4	358·8	63	N
54	L(B)	3	349	−3	61·4	180·7	180·7	79·5	267·0	9	N
55	L (B)	3	350	7	67·4	54·6	69·7	79·6	299·2	14	N
					(e) *Annobon*						
1	L (B)	3†	120	−49	173·2	56·7	173·2	25·2	129·7	9	—
2	L (B)	2†	197	14	209·1	71·9	342·1	69·3	310·0	13	R
3	L (B)	4†	173	17	39·3	425·3	1637·9	33·3	85·4	2	R
6	L (B)	4†	171	19	27·7	4·8	28·2	71·1	160·4	17	R
7	L (B)	2†	224	2	74·1	32·8	170·6	45·4	276·1	19	R
8	L (B)	2†	180	7	1351·6	868·3	1583·8	87·4	352·1	6	R
9	L (B)	2†	334	46	24·3	89·3	454·9	52·0	326·9	11	N
10	L (B)	5†	191	−4	7·9	6·7	7·9	78·0	259·1	30	R
11	L (B)	2†	184	19	35·5	258·9	8436·1	80·1	339·0	2	R
12	L (B)	2†	351	−49	4004·8	2027·3	7624·3	59·3	200·1	2	N
13	L (B)	3†	331	−36	18·9	11·1	25·7	55·6	238·9	24	N
14	L (B)	3†	337	−60	30·5	22·9	62·9	44·3	209·1	15	N
15	L (B)	3†	323	−42	350·9	131·4	531·4	47·6	238·4	5	N
17	L (B)	3†	330	−48	149·7	454·2	1900·0	49·9	226·9	2	N
18	L (B)	3†	337	−38	312·7	77·5	312·7	59·7	230·6	7	N
19	L (B)	2†	347	−30	140·6	242·4	424·5	70·6	223·8	12	N
20	I (T)	2†	152	−53	1·9	638·3	7942·8	46·0	151·9	2	—

†Annobon samples are taken from one or two lump samples from each site, L = lava, I = intrusive, B = basalt, T = trachyte, P = Phonolite or tephrite

Table 3

Gulf of Guinea volcanic province, group analysis

		D	I	No. of sites	R	k	α_{95}	Palaeomagnetic Pole Position °E	°N
Cameroun (9·5° E, 4·5° N)		4·6	4·3	32	30·29	18·1	6·1	121·8 ($d\psi = 3·1°$	85·1 $d\chi = 6·2°$)
Fernando Poo (9° E, 3·5° N)		359·9	−3·0	52	49·53	206·8	4·4	189·0 ($d\psi = 2·2°$	85·0 $d\chi = 4·5°$)
Principe (7·5° E, 1·5° N)		187·2	−3·2	25	22·90	11·4	8·9	276·6 ($d\psi = 4·5°$	−82·8 $d\chi = 9·0°$)
São Tomé (6·5° E, 0° S)		359·2	−7·0	49	46·51	192·9	4·7	199·4 ($d\psi = 2·4°$	86·4 $d\chi = 4·8°$)
Annobon (5·5° E, 1° S)	(i)	189·2	11·2	7	6·64	16·5	15·3	302·8 ($d\psi = 7·9°$	−79·7 $d\chi = 15·5°$)
	(ii)	337·1	−44·3	7	6·86	43·0	9·3	225·0 ($d\psi = 7·3°$	56·7 $d\chi = 11·7°$)

given K–Ar age dates of 2·8 and 2·4 My with a preferred average of 2·6 My. The anomalous pole position may have resulted from this lava being erupted during a magnetic field reversal and the age dates suggest that it might represent the Kaena reversed event of the Gauss epoch.

3.2.2 *São Tomé.* At least two polarity epochs are represented on this island. Along the northern coast an older normal succession of palagonite tuffs and pillow lavas formed by subaqueous eruption is overlain by younger reversed subaerial lavas. Along the east and north-east coast most of the lava units sampled have normal polarity but in two places reversed horizons have been found (sites 50 and 51). Hedburgh (1968) has obtained four age dates of 0·35 My and younger from lavas in this part of the island and it is inferred that the normally magnetized sites belong to the Bruhnes epoch. The sites taken in the south-east part of the island (34–48) may be somewhat older in age and Hedburg (1968) has formed ages ranging from 2·5–4·0 My for intrusions in this part of the island. Sites 35 and 43 are reversely magnetized and the remainder are normally magnetized. The sites 47 and 48 are from lavas which are morphologically young and are associated with recent cinder cones; it is inferred therefore that they belong to the Bruhnes epoch.

The palaeomagnetic collection from this island is believed to belong entirely to the last few million years although the basement rocks may be much older. Sites 7 and 24 give anomalous directions and sites 14, 48 and 53 are not significant, also there is only one sample from site 15 so these sites have been excluded from the final analysis. Taking the antiparallel direction of the 19 reversed sites, the remainder give a combined direction of $D = 359{\cdot}2°$, $I = 7{\cdot}0°$ ($\alpha_{55} = 4{\cdot}7°$).

3.2.3 *Principe.* Both normal and reversed lavas occur in the succession on this island and changes from normal to reversed polarity in the successions samples near the east, west and north-west coasts. Normally magnetized (sites 24 and 28) and reversely magnetized (site 25) intrusions are present in the west of the island and a reversely magnetized intrusion (site 16) is present in the centre of the island suggesting that the lava pile was intruded by phonolites and tephrites in more than one phase. Sites 4, 5, 7, 22 and 38 were insignificant at the 95 per cent confidence level before and after cleaning and sites 2, 6, 14, 17, 18, 19, 24, 26, 34, 39 and 40 give anomalous directions which do not appear to fall into normal or reversed groups; these have been excluded from the final analysis, but the reason for the large number of anomalous sites on this island is not known. The age of this collection is also not known precisely. The only age available is a K–Ar determination on a basalt flow of 24 My (Hedburg 1968).

3.2.5 *Fernando Poo.* The sites on this island were taken largely from the lower slopes of the Santa Isabel volcano and from the northern slopes of the San Carlos and San Jaoquin volcanoes (Fig. 5). All sites except numbers 50 and 51 are normally magnetized and since many of the flows are clearly very young, it is believed that virtually the whole volcanic edifice of this island has grown in Bruhnes epoch time and is thus younger than 0·7 My. This conclusion is consistent with a radiometric age obtained from the island by Hedburg (1968) although he has also reported one slightly older age. Doell & Cox (1961) have similarly shown that much of the island of Hawaii was built up during the present polarity epoch, and the possibility of the rapid accumulation of lavas on volcanic islands has been suggested elsewhere (Watkins, Richardson & Mason 1968).

Sites 33 and 48 are not significant and site 53 has an anomalous direction of magnetization. The combined direction obtained from the remainder of the sites and reversing sites 50 and 51 is $D = 359{\cdot}9°$, $I = -3{\cdot}0°$ ($\alpha_{95} = 4{\cdot}4°$).

3.2.6 *Cameroun.* All of the sites samples around the flanks of Mt Cameroun are normally magnetized. Hedburg (1968) has reported ages of 0·5 and 0·8 My from the Tiko area on the south-east flanks of this mountain and like the volcanoes of Fernando Poo, it is believed that the entire edifice of this volcano has been constructed in Bruhnes epoch times. Site 13 is taken on a lava which erupted from the western flank of Mt Cameroun in 1922.

Sites taken from the young volcanic area stretching to the north-east of Mt Cameroun are also entirely normally magnetized, and may belong to the Bruhnes epoch although no radiometric ages are available to confirm this. Sites 20, 21, 32–34, 41 and 42 are taken from areas of older basalts and sites 21 and 32 are reversely magnetized. Sites 7, 11, 30, 40 and 41 are not significant and have been excluded from the final analysis together with sites 10 and 43 for which only one sample was available. Sites 35, 36, 38 and 39 were taken from lava flows in one of the trachyte complexes in the Bamenda Highlands; sites 36 and 37 give anomalous directions and have been excluded from the final analysis and sites 37 and 39 are normally magnetized. The acceptable sites from the volcanic areas of Cameroun combine to give a mean direction of $D = 4{\cdot}6°$, $I = 4{\cdot}3°$ ($\alpha_{95} = 6{\cdot}1°$).

4. Palaeomagnetic poles

The pole position obtained from the younger lavas of Annobon is excluded from this discussion since it is anomalous for the young age of these lavas, and may represent a partial reversal of the magnetic field. Of the remaining pole positions those from São Tomé, Fernando Poo and Cameroun are derived from material Lower Miocene or younger in age, and the bulk of the collections from Fernando Poo and Cameroun belong to the Bruhnes epoch. The older pole position from Annobon is lower Miocene and the pole position from Principe is probably of similar age. This study has thus provided five new palaeomagnetic poles for the African plate in Upper Tertiary times.

The palaeomagnetic poles are without exception on the side of the geographic pole farthest from the sampling area confirming the observation of Wilson (1970) that this is generally true for Upper Tertiary times. The poles from São Tomé and Cameroun are practically on the line of longitude of the sampling site but those from Cameroun, Principe and Annobon are displaced to the right. Wilson (1970) has explained the displacement of the palaeomagnetic poles to the far side of the globe to the sampling site in terms of an off-centre displacement of the effective dipole source. It is interesting to note that the palaeomagnetic pole for Fernando Poo is significantly different from the present pole although it is derived from material which is almost entirely younger than 700 000 years old. It is clear therefore that the magnetic field at Fernando Poo has not approximated to a geocentric dipole field during this time, unless by some accident the palaeomagnetic sample is not truely representative of the magnetic field behaviour here.

It remains to be examined whether we can separate the effects of non-dipole behaviour and real polar wandering or continental drift in the Upper Tertiary. Palaeomagnetic poles obtained from the continental and oceanic portions of the African plate from Upper Tertiary rocks are shown in Fig. 10. Poles obtained from the younger lavas of the East African rift R1 (0–2·5 My) and R2 (2.5–5·0 My, Reilly 1970) are not significantly different from the present geographic pole. The pole R3 (5·0–7·5 My) shows a significant difference from the present pole, and is similar to poles obtained from the Cape Verde Islands (Watkins *et al.* 1968), Madiera (Watkins *et al.* 1966b), Tenerife (Watkins *et al.* 1966a) and the Turkana lavas of Kenya (Raja, Reilly & Mussett 1966). The pole from Madiera is Pliocene in age and that from Kenya is probably Middle Miocene. However, these poles are displaced at a distance from the present pole similar to the poles from São Tomé, Fernando Poo

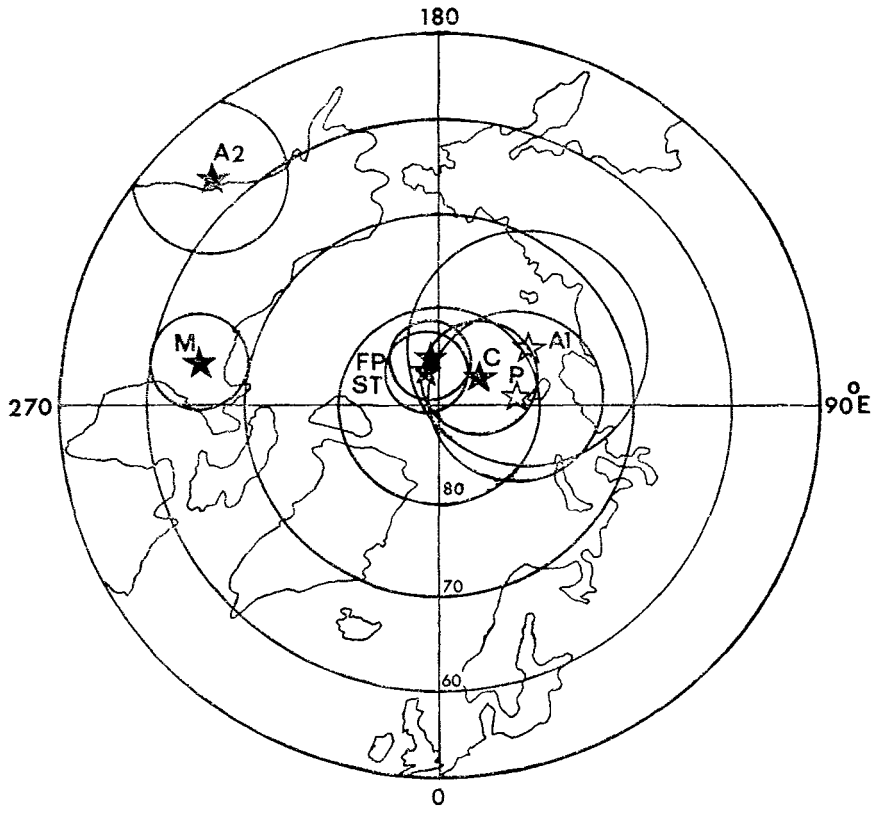


FIG. 9. Summary of palaeomagnetic poles falling in the Northern Hemisphere and 95 per cent confidence circles from the Gulf of Guinea volcanic province. The poles are: Cameroun (C), Fernando Poo (FP), Principe (P), São Tomé (ST), older lavas and intrusions of Annobon (A1), and young lavas of Annobon (A2). *M* is the mean Mesozoic pole of McElhinny *et al.* (1968).

and Cameroun, and since these latter poles are derived from very young material this displacement is likely to be due to a non-dipole complication rather than real polar shift or continental drift. The pole for the Ethiopian traps (Brock, Gibson & Gacci 1970) was originally thought to be Eocene–Oligocene but new age results suggest that it is no older than Lower Miocene (Rex, Gibson & Dakin 1971) and approximately the same age as the poles from Annobon and Principe. These poles are displaced 7–10° from the present pole and might represent a real polar shift/continental movement of a few degrees in the Lower Miocene–Pliocene interval. However, even this conclusion is tentative since the 95 per cent confidence circles of the Principe and Annobon poles include the present pole. The pole from Hierro (Watkins *et al.* 1966a) is close to that of Principe although the former pole is believed to be derived entirely from Bruhnes epoch age lavas. The other poles from the Canary Islands are of greater age but unlikely to be older than Eocene (Watkins *et al.* 1966a); until ages for these islands are forthcoming it is not possible to decide whether the displacement of the poles from Gomera, Gran Canaria and Lanzarote from the present pole can be interpreted in terms of plate movement.

There is therefore as yet no palaeomagnetic reason for believing that the African plate has moved relative to the pole in Upper Tertiary times. The nine Mesozoic

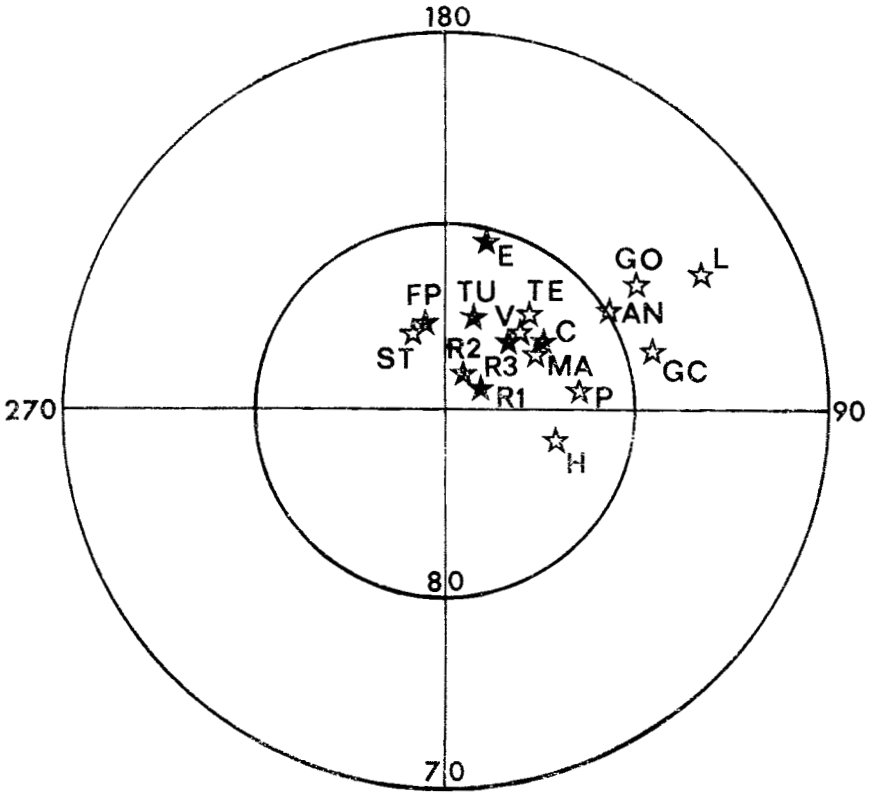


FIG. 10. Summary of palaeomagnetic north pole positions from the continental (closed stars) and oceanic (open stars) parts of the African plate. The continental poles are: Fernando Poo (FP), Ethiopian traps (E), Turkana lava (TU), lavas of East African rift valley: 0–2·5 My (R1), 2·5–5·0 My (R2), 5·0–7·5 My (R3), and Cameroun (C). The oceanic poles are: São Tomé (ST), Principe (P), Annobon (AN), Cape Verde Islands (V), Madiera (MA), Gran Canaria (GC), Gomera (GO), Hierro (H), Tenerife (TE) and La Palma (L).

poles from Africa group closely at near 65° N and 261° E (McElhinny *et al.* 1968) and the mean pole is shown as *M* in Fig. 9. The youngest Mesozoic pole is about 100 My in age and the shift from *M* to the Upper Tertiary pole positions must have taken place at some time in the interval 100–25 My B.P.

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